

Covering systems with moduli of the form $p - 1$, $p \geq 5$ prime

Erdős Problem #273

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Abstract

Abstract to be written once the first verified result has landed; the introduction below states the problem.

1 Introduction

A *covering system* of the integers is a finite collection of arithmetic progressions

$$\{a_i \pmod{m_i}\}_{i=1}^k, \quad m_i \geq 2,$$

whose union is $\mathbb{Z}$; equivalently, every integer lies in at least one residue class $a_i \pmod{m_i}$. Covering systems were introduced by Erdős in 1950, and a sequence of problems concerning their structure have driven a substantial body of work in combinatorial number theory. The present manuscript concerns one of these problems, recorded as Problem #273 in the Erdős–Graham survey [1].

Conjecture 1 (Erdős #273). *Does there exist a covering system $\{a_i \pmod{m_i}\}_{i=1}^k$ of $\mathbb{Z}$ such that each modulus m_i has the form $m_i = p_i - 1$ for some prime $p_i \geq 5$?*

Write $\mathcal{M} := \{p-1 : p \text{ prime}, p \geq 5\} = \{4, 6, 10, 12, 16, 18, 22, 28, 30, 36, 40, 42, 46, 52, 58, 60, \dots\}$ for the set of allowed moduli. The question is whether $\mathbb{Z}$ admits a finite covering whose moduli are drawn from $\mathcal{M}$.

The restriction $p \geq 5$ is essential. Selfridge observed that allowing $p = 3$ admits a covering system using divisors of $360 = \text{lcm}(2, 3, 4, 5, 6, 8, 9, 10, 12)$, all of which are of the form $p - 1$ for $p \in \{3, 5, 7\}$. The Selfridge construction crucially uses the modulus $2 = 3 - 1$. Forbidding $p = 3$ eliminates the modulus 2 entirely and changes the problem: any covering by progressions modulo elements of $\mathcal{M}$ must account for every parity class without ever using the all-purpose modulus 2, and the second-smallest available modulus is 4.

Two pieces of recent machinery bear on the question. Hough [2] resolved the longstanding Erdős minimum-modulus problem by establishing a finite upper bound on the smallest modulus required in any covering system, ruling out covering systems with all moduli larger than a fixed constant. Balister, Bollobás, Morris and Sahasrabudhe [3] extended Hough’s distortion method to a general non-existence framework for covering systems whose moduli are constrained to a multiplicatively structured set. Specializing either machine to $\mathcal{M}$ is a natural avenue, although neither result has been adapted to the present restriction in the published literature.

To the author’s knowledge, Conjecture 1 is open as of 2026; no partial result—a lower or upper bound on the smallest required modulus, on the number of progressions, or on the least common multiple of any putative covering—appears in the published literature.

The main result of this manuscript is recorded in Section 2, currently in preparation.

2 Main result

Statement and proof in preparation.

References

- [1] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, L'Enseignement Mathématique **28** (1980), Problem #273, p. 24.
- [2] B. Hough, *Solution of the minimum modulus problem for covering systems*, Annals of Mathematics **181** (2015), no. 1, 361–382. arXiv:1307.0874.
- [3] P. Balister, B. Bollobás, R. Morris, and J. Sahasrabudhe, *On the Erdős covering problem: the density of the uncovered set*, Inventiones Mathematicae **228** (2022), no. 1, 377–414. arXiv:1811.03547.